CM 1111

FUNDAMENTALS OF MATHEMATICS

HONOURS DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

BATCH 25- LEVEL 1 (SEMESTER I)

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CHAPTER 04

LIMITS AND DERIVATIVES

LECTURE 07

LIMITS

HOW LIMITS ARE USEFUL FOR IT STUDENTS

➢ Algorithm Design: Big-O notation.

➢ Hardware/Software: Floating-point precision.

➢ Data Science: Model convergence.

➢ Graphics: Anti-aliasing.

➢ Networking: Bandwidth optimization.

➢ Data Structures (Hashing & Load Factors) ➢ Machine Learning (Gradient Descent)

A NUMERICAL & GRAPHICAL APPROACH TO LIMIT

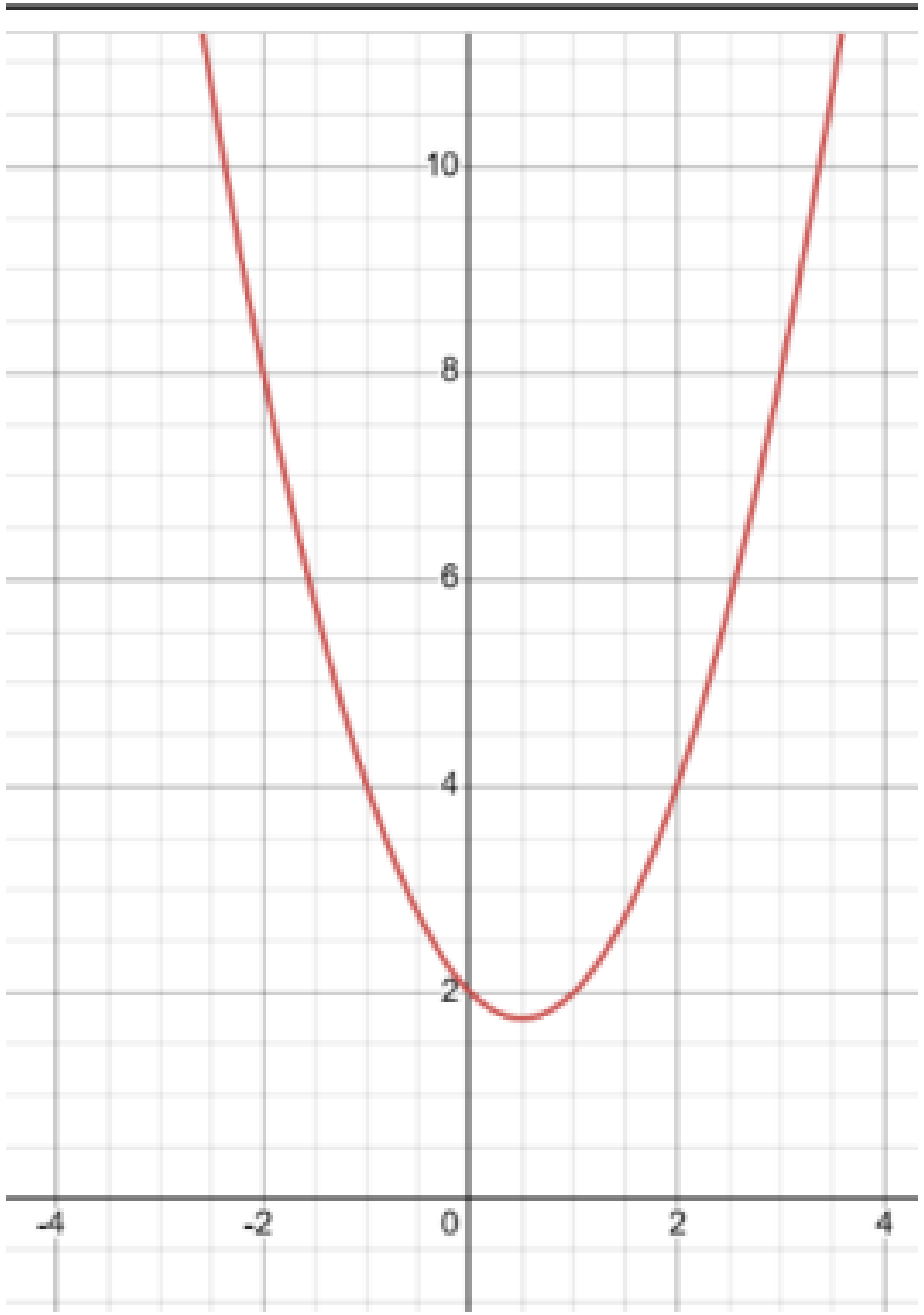
INTRODUCTION TO THE CONCEPT OF LIMITS

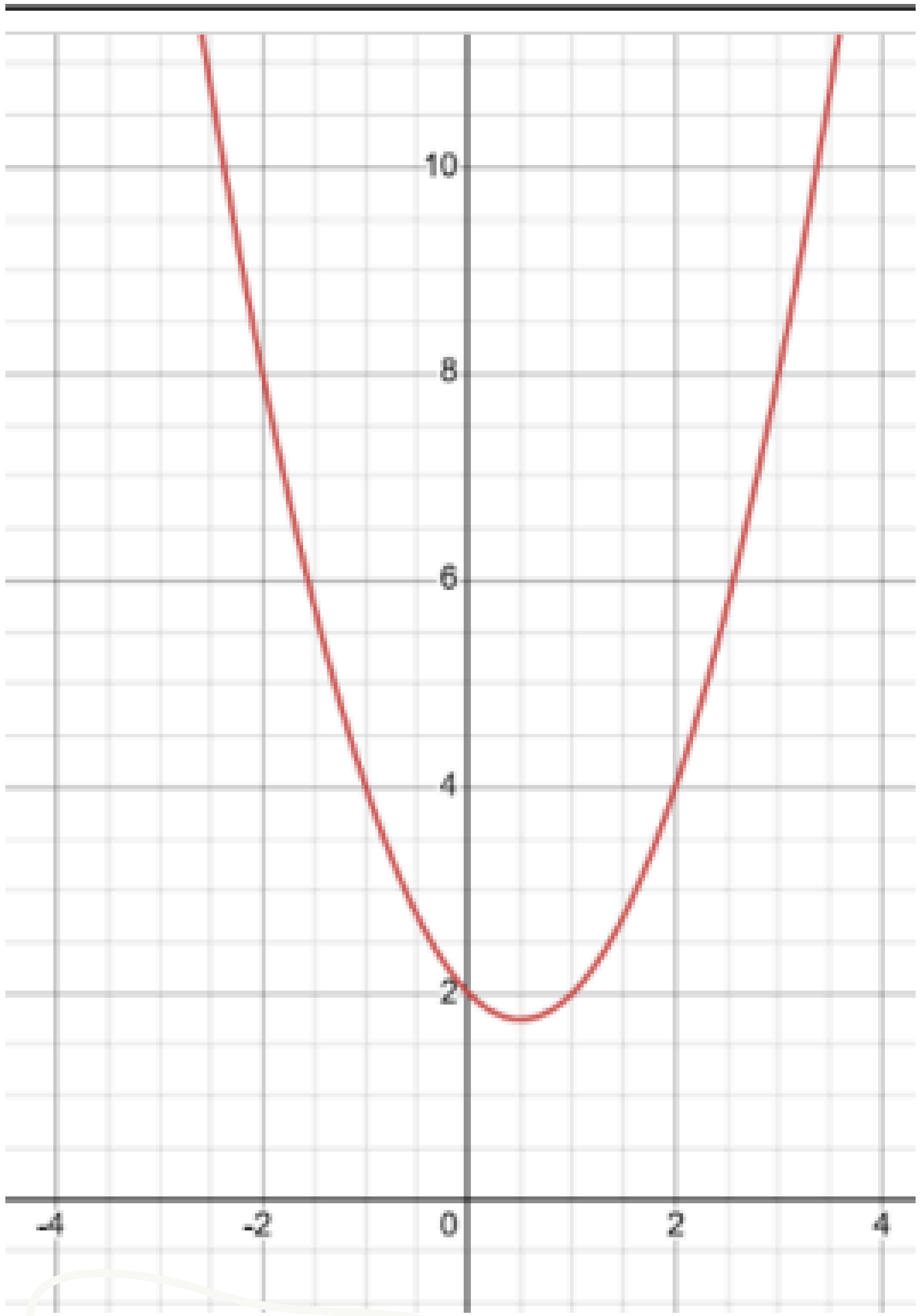
- Calculus is the study of change, and limits are its foundational building blocks.
- How do a function's outputs ($f(x)$) behave as its inputs (x) get closer and closer to a specific number, a?
- If the outputs approach a single, predictable number, L, as the inputs approach a, we call L the limit of the function at that point.

Let's investigate the behavior of the function f defined by $f(x) = x^2 - x + 2$ for values x of near 2. The following table gives values of $f(x)$ for values of x close to 2 but not equal to 2.

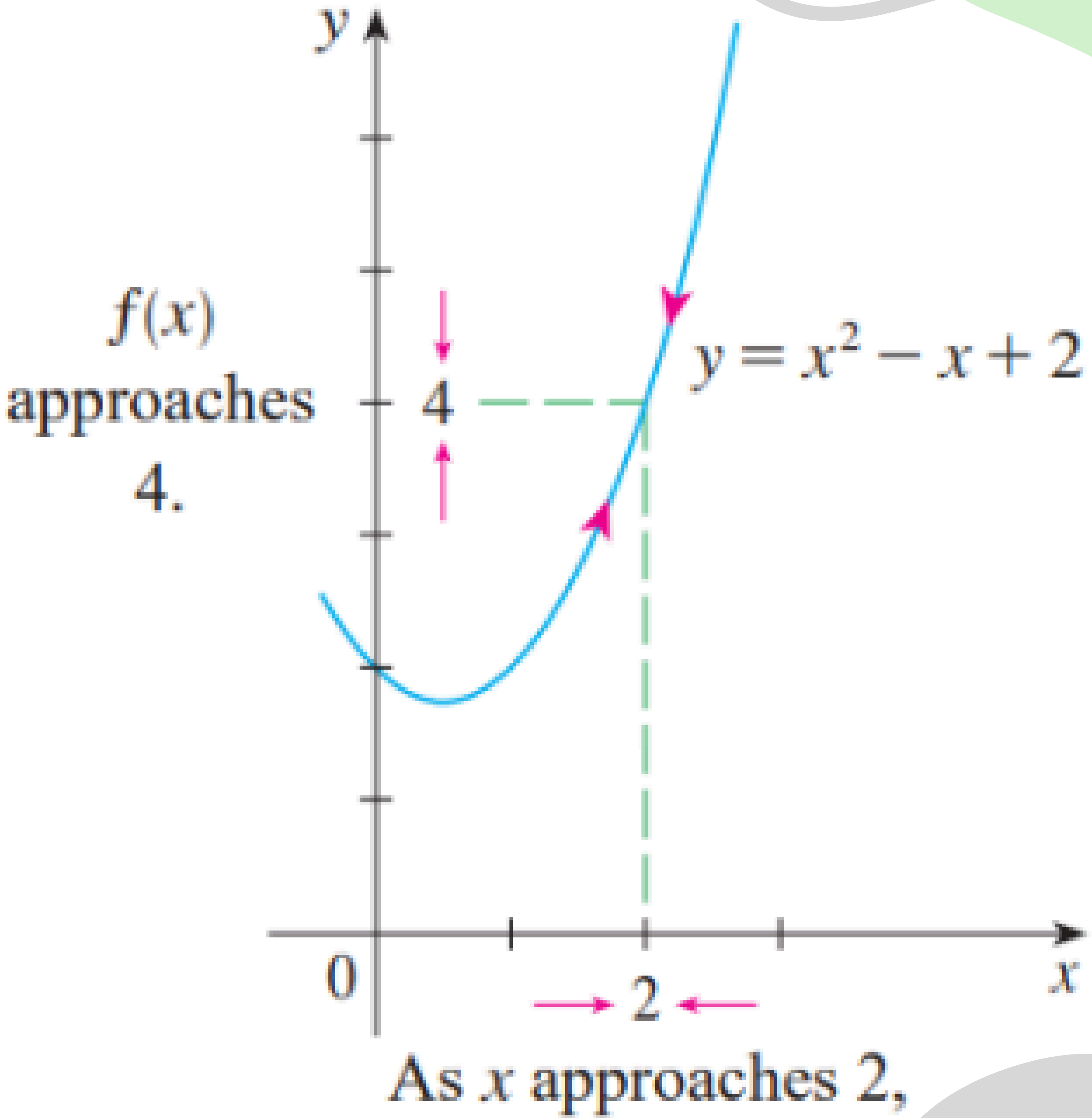
x	$f(x)$	x	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001

$f(x) = x^2 - x + 2$



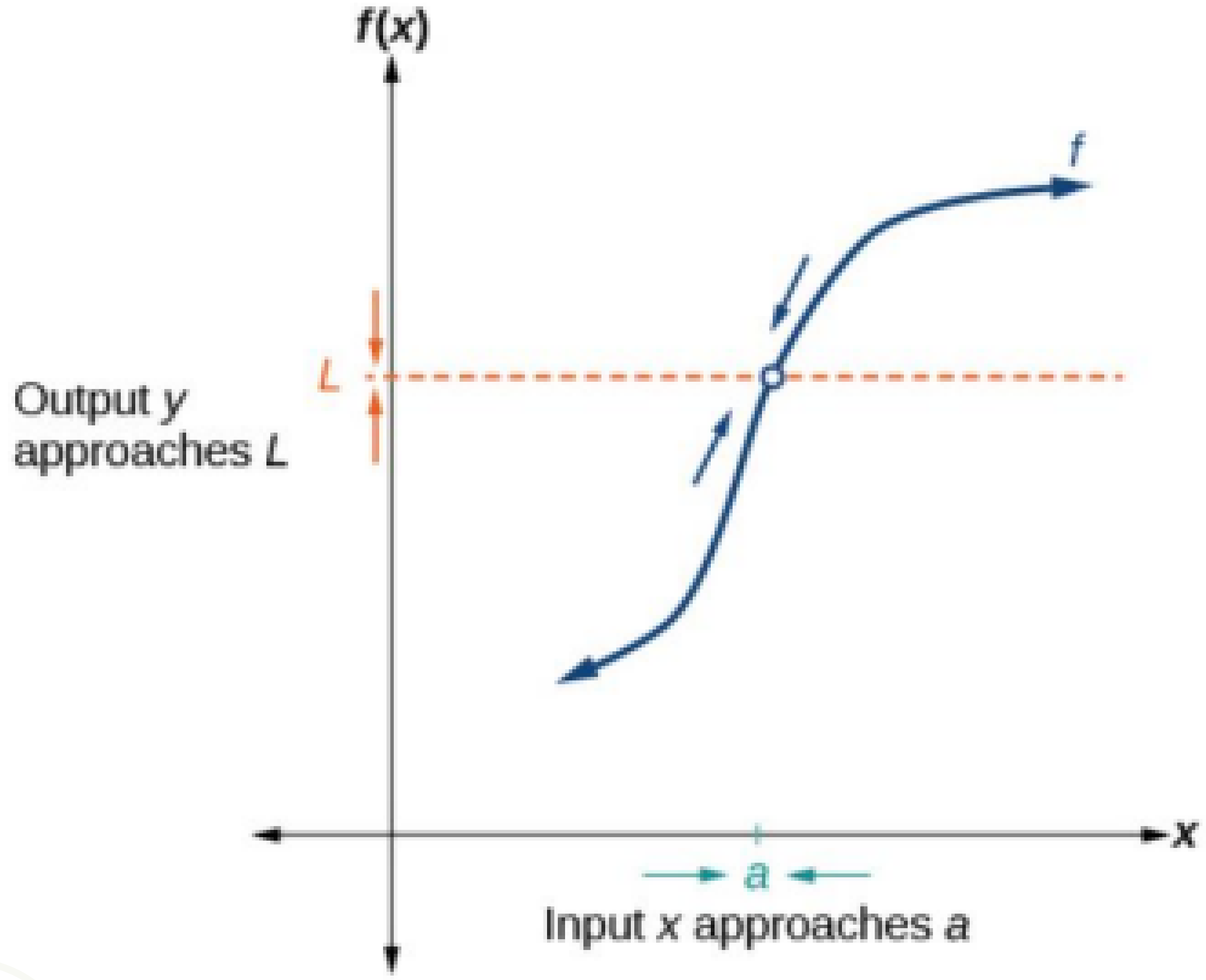


10
8
6
4
2
-4
-2
0
2
4



y
$f(x)$
approaches
4.
4
$y = x^2 - x + 2$
0
2
x
As x approaches 2,

Below figure provides a visual representation of the mathematical concept of limit



Exercise 01

Investigate the behavior of the function $f(x) = 5x + 3$ for values x of near 2 using numerical and grpahical approach

Exercise 01 - Solution

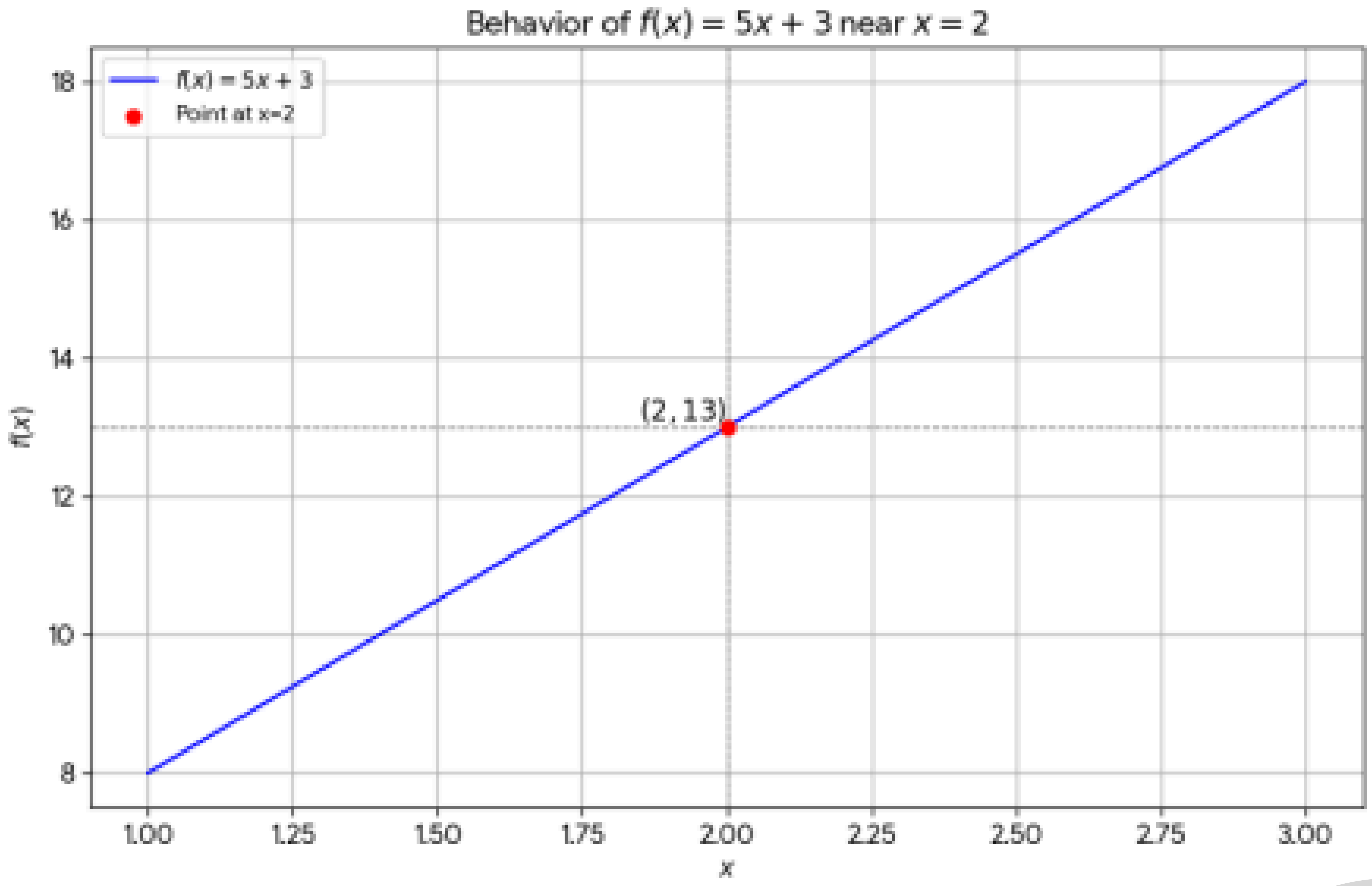
$f(x) = 5x + 3$

x	f(x)
1.900	12.500
1.990	12.950
1.999	12.995

1

2.001	13.005
2.010	13.050
2.100	13.500

2



DEFINITION - LIMITS

Definition Suppose $f(x)$ is defined when x is near the number a. (This means that f is defined on some open interval that contains a, except possibly at a itself.) Then we write

$$\lim_{x \to a} f(x) = L$$

and say "the limit of $f(x)$, as x approaches a, equals L"

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a.

THEOREM

As x approaches a, the limit of $f(x)$ is L if the limit from the left exists and the limit from the right exists and both limits are L. That is,

if $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = L$, then $\lim_{x\to a} f(x) = L$.

REMARKS

- When we write $\lim_{x \to a} f(x)$ we are indicating that x is approaching a from both sides.
- If we want to be specific about the side from which the x-values approach the value a, we use the following notations:

$\lim_{x \to a^-} f(x)$ to indicate the limit from the left (that is, where $x < a$).

$\lim_{x \to a^+} f(x)$ to indicate the limit from the right (that is, where $x > a$).

- These are called left-hand and right-hand limits, respectively.
- In order for **a limit to exist, both the left-hand and right-hand limits must exist and be the same.**

Exercise 02

Let $f(x)$ =

$$\frac{x-1}{x^2-1}$$

a. What is $f(1)$?

b. What is the value of $f(x)$ as x approach 1?

Exercise 02 - Solution

To find $f(1)$, we substitute $x = 1$ into the function:

$$f(1) = \frac{1-1}{1^2-1} = \frac{0}{0}$$

Since division by zero is **undefined**, f(1) does not exist

Exercise 02 - Solution

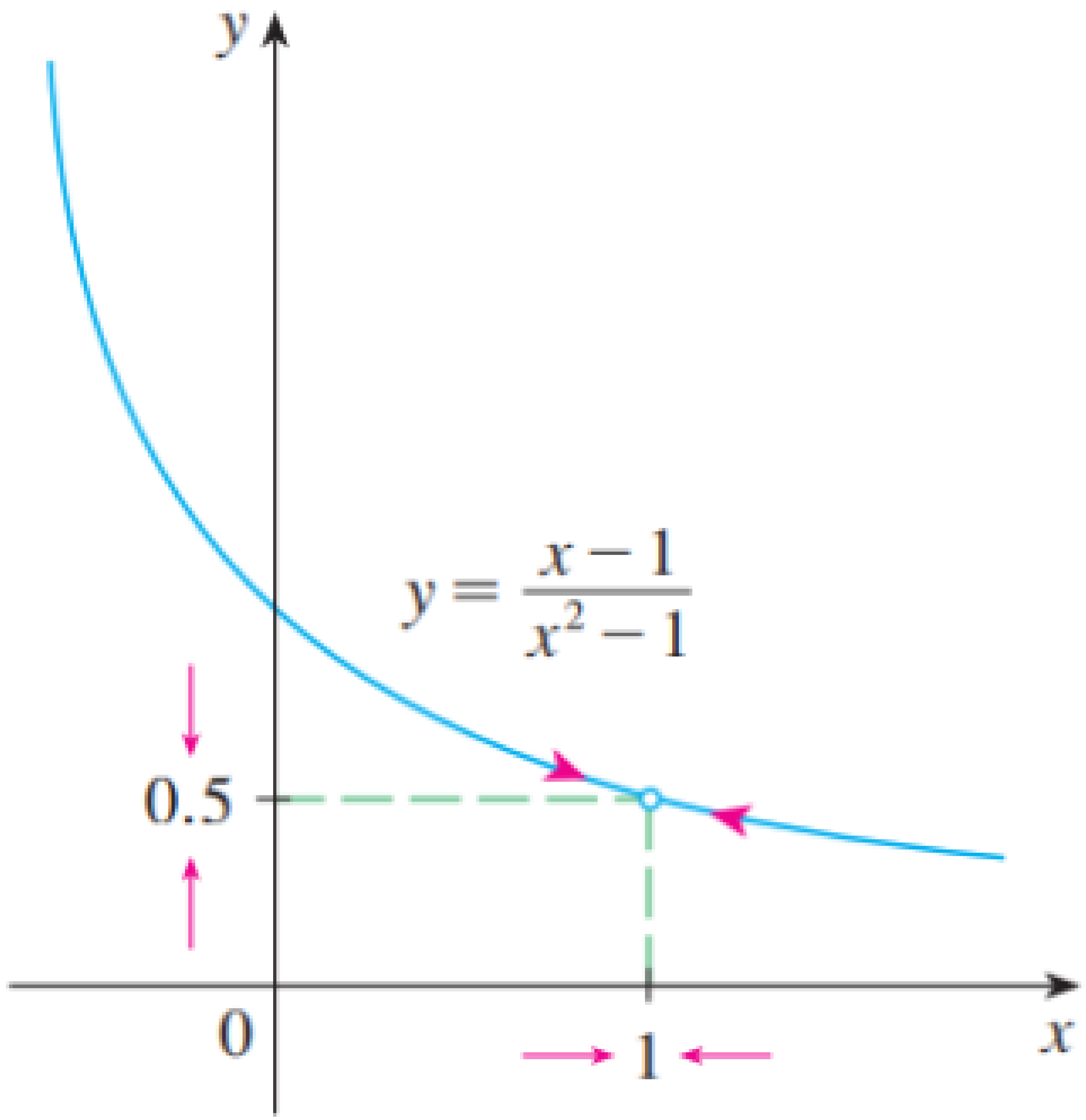
$x < 1$	$f(x)$
0.5	0.666667
0.9	0.526316
0.99	0.502513
0.999	0.500250
0.9999	0.500025

$x > 1$	$f(x)$
1.5	0.400000
1.1	0.476190
1.01	0.497512
1.001	0.499750
1.0001	0.499975

On the basis of the values in the tables, we make the guess that

$$\lim_{x \to 1} f(x) = 0.5$$

Exercise 02 - Solution



Exercise 03 – Limits of a piecewise function

$$\text{Let } f(x) = \begin{cases} -2x+1 & x<0 \\ x-1 & x\geq 0 \end{cases}$$

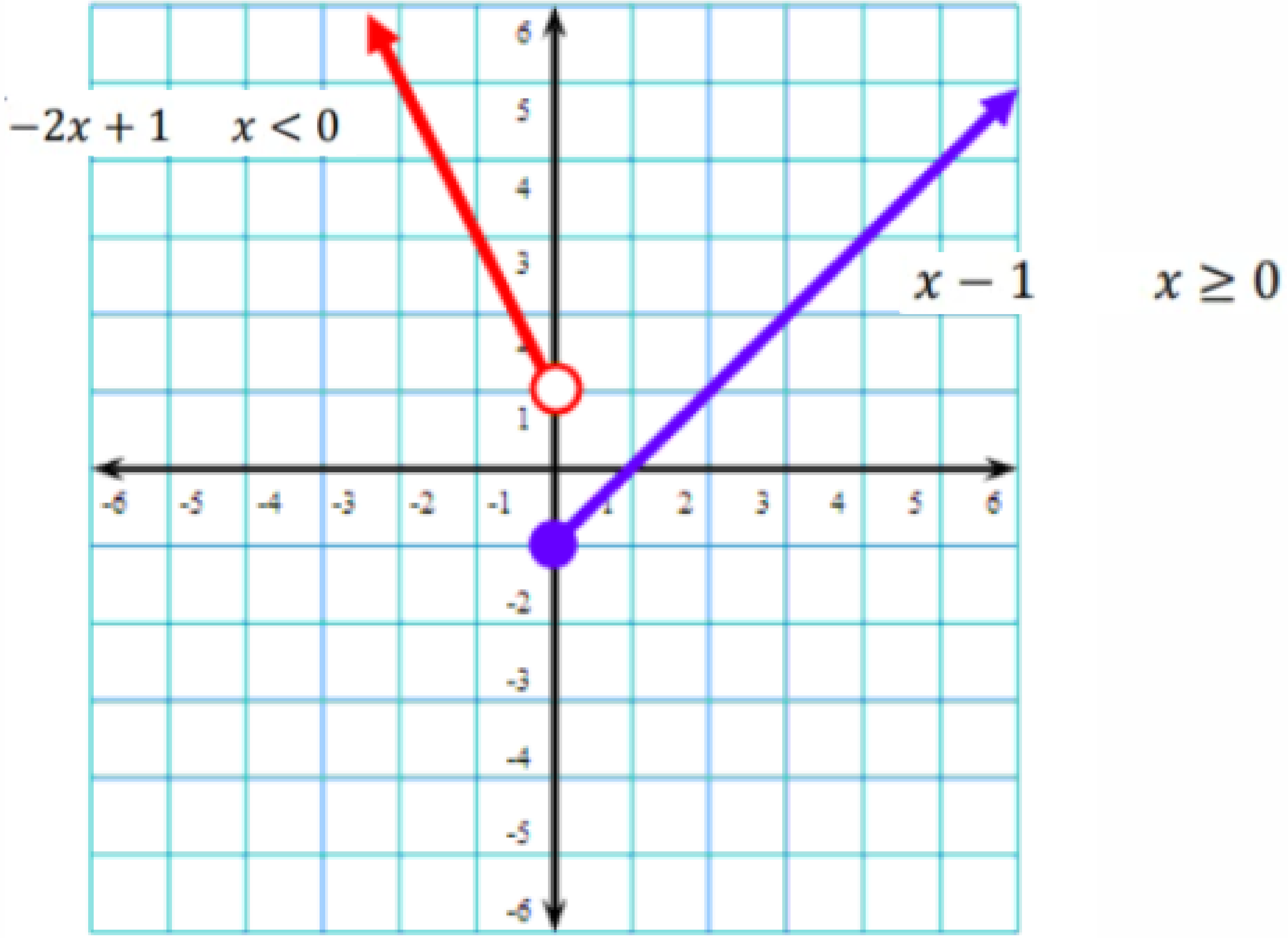
Graph the function and find each of the following limits, if they exist.

a) Find $\lim_{x\to -1} f(x)$

b) Find $\lim_{x\to +2} f(x)$

c) Find $\lim_{x\to 0} f(x)$

Exercise 03- Solution



Exercise 03- Solution

$$f(x) = -2x + 1$$

a) Find $\lim_{x \to -1} f(x)$

x	f(x)
-1.100	3.200
-1.010	3.020
-1.001	3.002
-0.999	2.998
-0.990	2.980
-0.900	2.800

$\lim_{x \to -1} f(x) = 3$

Exercise 03- Solution

$$f(x) = x - 1$$

b) Find $\lim_{x \to +2} f(x)$

x	f(x)
1.900	0.900
1.990	0.990
1.999	0.999
2.001	1.001
2.010	1.010
2.100	1.100

$\lim_{x \to +2} f(x) = 1$

Exercise 03- Solution

c) Find $\lim_{x \to 0} f(x)$

x	f(x)
-0.100	1.200
-0.010	1.020
-0.001	1.002

$f(x) = -2x + 1$

0.001	-0.999
0.010	-0.990
0.100	-0.900

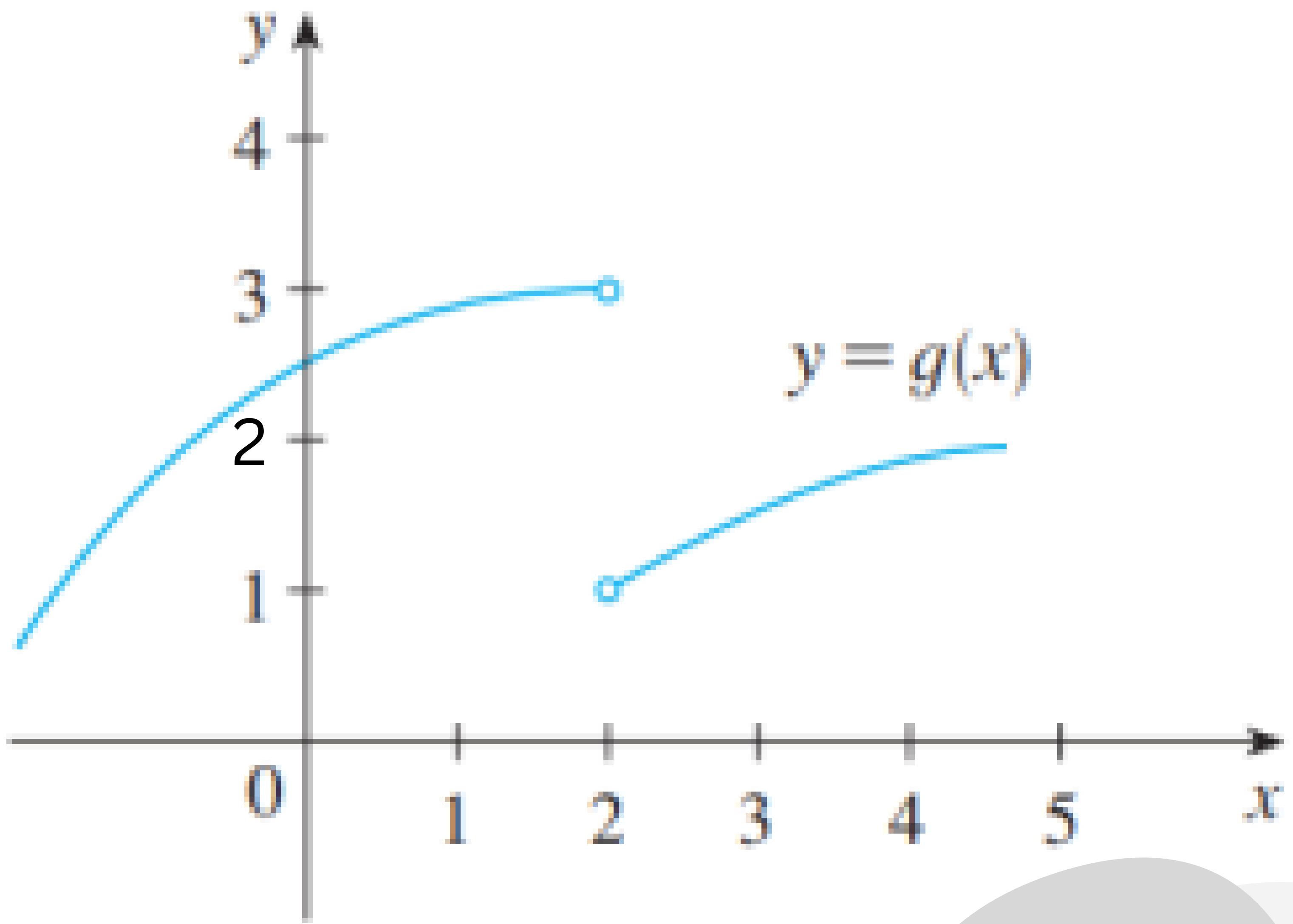
$f(x) = x - 1$

The limit does not exist.

Exercise 04

The graph of a function g is shown in Figure . Use it to state the values (if they exist) of the following:

(a) $\lim_{x \to 2^-} g(x)$ (b) $\lim_{x \to 2^+} g(x)$ (c) $\lim_{x \to 2} g(x)$



Exercise 05 - Limits Involving Infinity

Let $f(x) = \frac{1}{x}$.

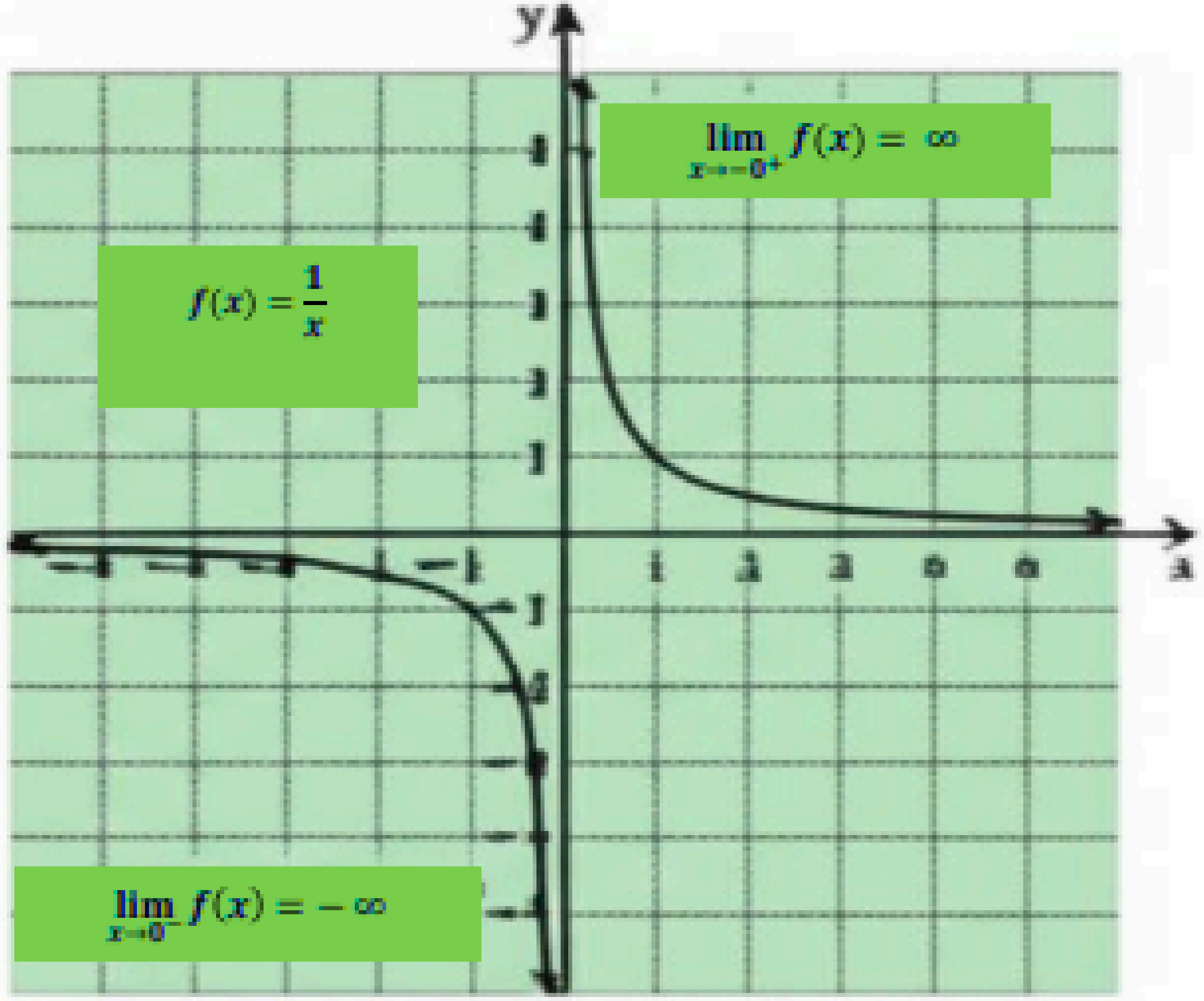
1. Find $\lim_{x \to 0^-} f(x)$.
2. Find $\lim_{x \to 0^+} f(x)$.
3. Use the information from parts (1) and (2) to form a conclusion about $\lim_{x \to 0} f(x)$.

Limit Numerically

$\lim_{x \to 0^-} f(x)$	$f(x)$
-0.1	-10
-0.01	-100
-0.001	$-1{,}000$
-0.0001	$-10{,}000$

$\lim_{x \to 0^+} f(x)$	$f(x)$
0.1	10
0.01	100
0.001	1,000
0.0001	10,000

Limit Graphically



$$\lim_{x \to 0^-} f(x). = -\infty$$

$$\lim_{x \to 0^+} f(x). = +\infty$$

$-\infty \neq +\infty$, the limit as $x \to 0$ does not exist.

Exercise 06

Let $f(x) = \dfrac{1}{x}$.

Find $\lim_{x \to \infty} f(x)$ and $\lim_{x \to -\infty} f(x)$.

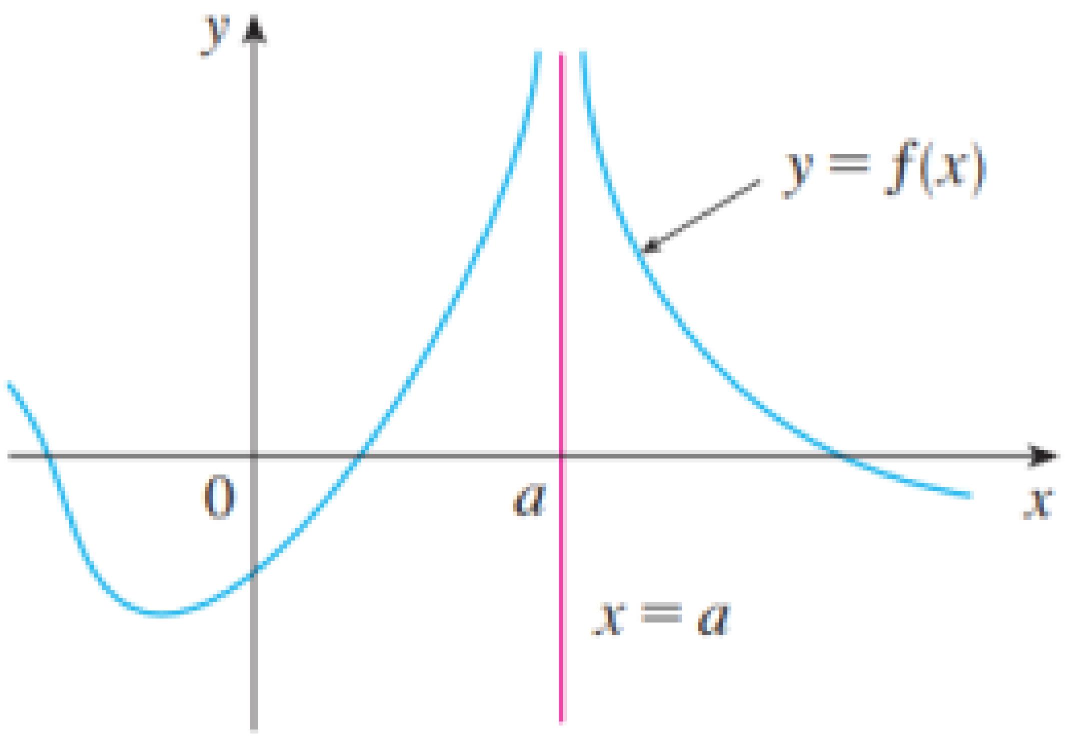


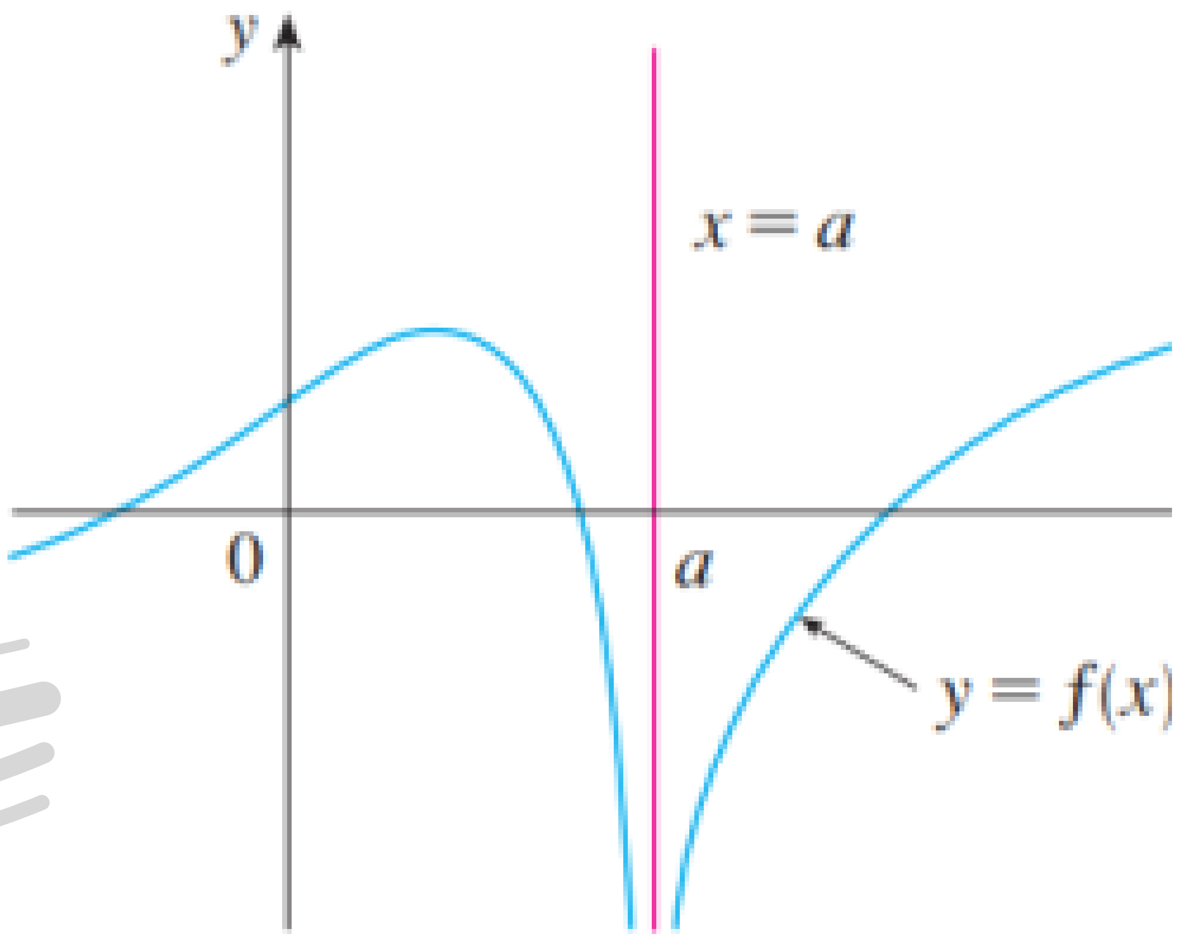
FIGURE
$\lim_{x \to a} f(x) = \infty$

Definition Let f be a function defined on both sides of a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a, but not equal to a.

A similar sort of limit, for functions that become large negative as gets close to , is defined



FIGURE

$\lim_{x \to a} f(x) = -\infty$

Definition Let f be defined on both sides of a, except possibly at a itself. Then

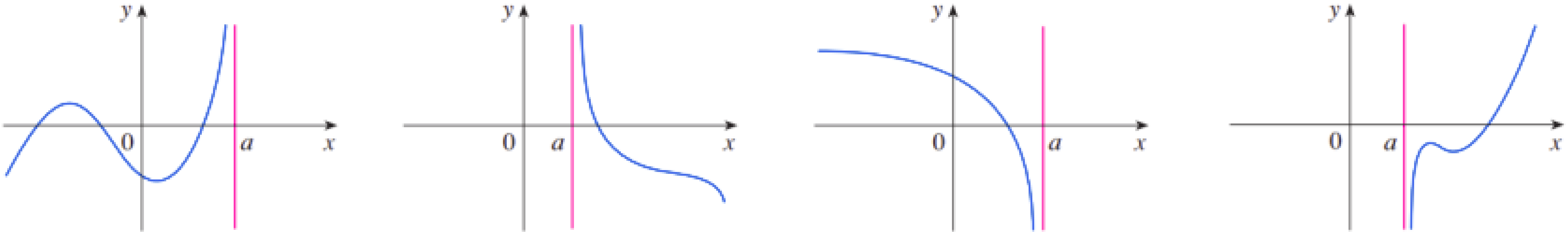
$$\lim_{x \to a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.

Definition The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a} f(x) = -\infty \qquad \lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$



(a) $\lim_{x \to a^-} f(x) = \infty$ (b) $\lim_{x \to a^+} f(x) = \infty$ (c) $\lim_{x \to a^-} f(x) = -\infty$ (d) $\lim_{x \to a^+} f(x) = -\infty$

Exercise 07

Consider the function f given by $f(x) = \frac{1}{x-3} + 4$

Find each of the following limits, if they exist. If necessary, state that the limit does not exist.

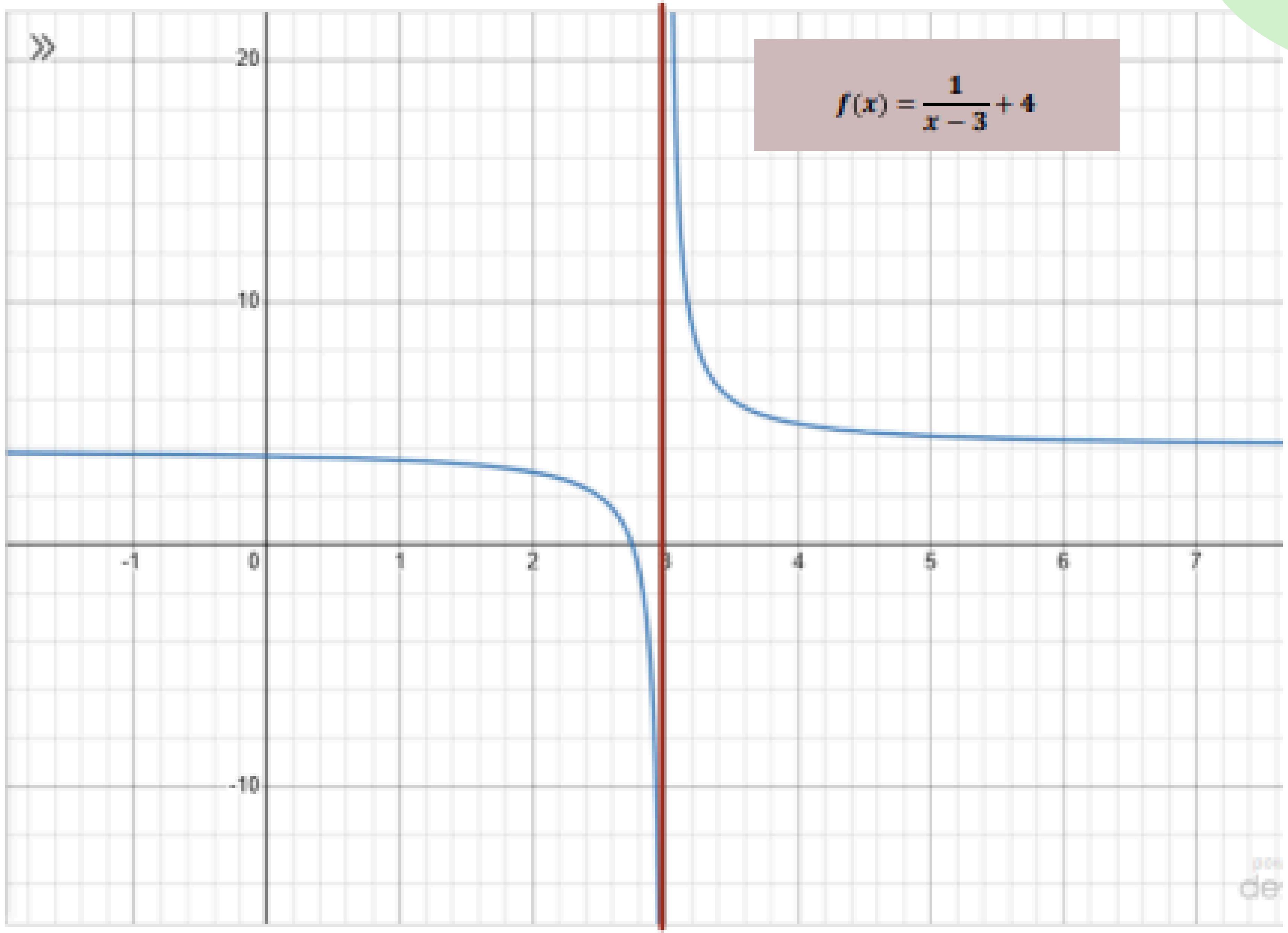
a) $\lim_{x \to 3} f(x)$

b) $\lim_{x \to 4} f(x)$

c) $\lim_{x \to \infty} f(x)$

a) $\lim_{x \to 3} f(x)$

x	f(x)
2.900	-6.0
2.990	-96.0
2.999	-996.0
3.001	1004.0
3.010	104.0
3.100	14.0



Since the left-hand and right-hand limits are not equal, the limit at x=3 does not exist.

b) $\lim_{x \to 4} f(x)$

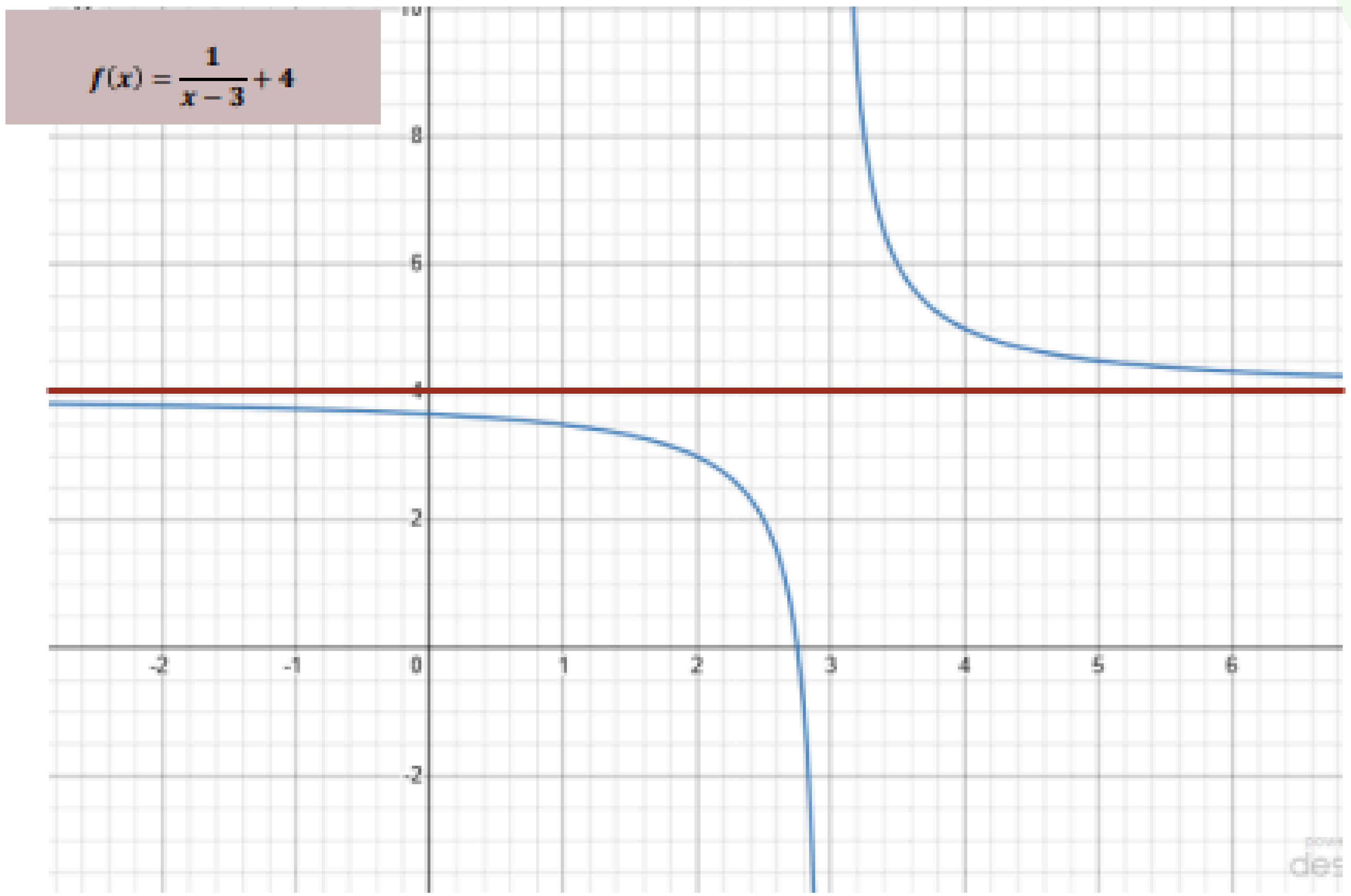
x	f(x)
3.900	5.111111
3.990	5.010101
3.999	5.001001

4.001	4.999001
4.010	4.990099
4.100	4.909091

As x approaches 4 from both sides, the value of f(x) gets closer to 5.

c) $\lim_{x \to \infty} f(x)$

x	f(x)
10	4.142857
100	4.010309
1000	4.001003
10000	4.000100
100000	4.000010



Take Home Activity

Estimate the value of $\lim_{t \to 0} \dfrac{\sqrt{t^2 + 9} - 3}{t^2}$.

Thank you